Dunedin Weather — Data Preparation

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Introduction

This project was undertaken to answer a practical question posed by a family member: what is the weather like in Dunedin, on average? The intended output is a single printed page of visualisations to be kept at the front door — a seasonal reference that allows a person leaving the house to quickly form expectations about temperature, wind, and rainfall without consulting a daily forecast. The design principle guiding this project is coverage over specificity. Rather than precise hourly predictions, the visualisations aim to represent typical conditions across seasons and months, from which a viewer can make practical inferences — whether to bring a rain jacket, a windbreaker, or neither. The most useful information the poster communicates is the expected temperature and wind direction and speed for the season a person finds themselves in. Historical data was chosen over point forecast data deliberately. A weather forecast describes what is predicted to happen on a specific future date. Historical climate data describes what has actually happened over many years — it is a more honest answer to the question "what is Dunedin weather like" than any single forecast could be. Beyond its practical purpose, this project serves as a portfolio piece demonstrating applied data science skills including data acquisition, preparation, exploratory analysis, and communication. The analysis is conducted in R and documented in R Markdown to ensure full reproducibility.

Data Description

Data were obtained from NIWA's DataHub (data.niwa.co.nz), New Zealand's national climate database. All data come from the Dunedin Musselburgh EWS station (agent no. 15752), an automatic weather station located at 170.5057°E, 45.8335°S in the Musselburgh area of Dunedin, near the harbour. The station record spans 8 August 1997 to 17 June 2026. Six variables were downloaded as separate CSV files, each containing hourly observations: mean temperature (°C), maximum and minimum temperature (°C), grass temperature (°C), relative humidity (%), wind speed (m/s) and direction (degrees true), gust speed (m/s) and direction (degrees true), rainfall (mm), atmospheric pressure (hPa), and sunshine hours. All variables are recorded at hourly resolution with a 1-hour preceding period. Pressure data begins 20 days after the temperature record (1997-08-28 versus 1997-08-08), and sunshine data begins in April 2002, reflecting different installation dates for those sensors. Prior to filtering, the full dataset comprises 252,519 hourly observations. Cloud cover was identified as a variable of interest during project planning but is not available from the NIWA DataHub for this station and has therefore been excluded from the analysis. The six files were joined into a single dataset on the UTC datetime column, then filtered to begin in 12 April 2002 to ensure all variables are present across a consistent time window. Timestamps were converted from UTC to New Zealand time. The final prepared dataset spans 12 April 2002 to June 2026. A decision on whether to further restrict the analysis to the most recent 10 years (2016–2026) has been deferred to the EDA stage, where a distributional comparison between earlier and later periods will be performed to assess whether recent climate conditions differ meaningfully from the earlier record.

```r
# --- Load libraries ---

library(tidyverse)
library(lubridate)
library(janitor)
```

```r
# --- Read and clean each file ---

temperature <- read_csv("Data/Raw/Temperature__hourly.csv") %>%
  clean_names() %>%
  select(
    datetime_utc = observation_time_utc,
    temp_mean    = mean_temperature_deg_c,
    temp_max     = maximum_temperature_deg_c,
    temp_min     = minimum_temperature_deg_c,
    temp_grass   = grass_temperature_deg_c,
    humidity     = mean_relative_humidity_percent
  )

wind <- read_csv("Data/Raw/Wind__hourly.csv") %>%
  clean_names() %>%
  select(
    datetime_utc   = observation_time_utc,
    wind_direction = direction_deg_t,
    wind_speed     = speed_m_s
  )

gust <- read_csv("Data/Raw/Gust__hourly.csv") %>%
  clean_names() %>%
  select(
    datetime_utc   = observation_time_utc,
    gust_direction = direction_deg_t,
    gust_speed     = speed_m_s
  )

rain <- read_csv("Data/Raw/Rain__hourly.csv") %>%
  clean_names() %>%
  select(
    datetime_utc = observation_time_utc,
    rainfall_mm  = rainfall_mm
  )

pressure <- read_csv("Data/Raw/Pressure__hourly.csv") %>%
  clean_names() %>%
  select(
    datetime_utc = observation_time_utc,
    pressure_msl = mean_sea_level_pressure_hpa,
    pressure_stn = station_level_pressure
  )

sunshine <- read_csv("Data/Raw/Sunshine__hourly.csv") %>%
  clean_names() %>%
  select(
    datetime_utc = observation_time_utc,
    sunshine_hrs = sunshine_hrs
  )
```

Before joining, each file was checked for duplicate timestamps using count(). No duplicates were found in any file, confirming that each dataset has a unique observation per hour. The files were then joined using a left join on datetime_utc, with temperature as the base file.”

```r
# --- Join all files on datetime ---

weather <- temperature %>%
  left_join(wind,     by = "datetime_utc") %>%
  left_join(gust,     by = "datetime_utc") %>%
  left_join(rain,     by = "datetime_utc") %>%
  left_join(pressure, by = "datetime_utc") %>%
  left_join(sunshine, by = "datetime_utc")

# --- Check the join result ---

nrow(weather)
```

```
## [1] 252519
```

```r
glimpse(weather)
```

```
## Rows: 252,519
## Columns: 14
## $ datetime_utc   <dttm> 1997-08-08 01:00:00, 1997-08-08 02:00:00, 1997-08-08 0~
## $ temp_mean      <dbl> 8.5, 8.9, 8.6, 8.3, 8.1, 7.1, 6.7, 6.4, 6.0, 6.0, 5.9, ~
## $ temp_max       <dbl> 9.1, 9.1, 9.0, 8.4, 8.4, 7.9, 6.9, 6.6, 6.3, 6.1, 6.1, ~
## $ temp_min       <dbl> 8.1, 8.7, 8.2, 8.1, 7.8, 6.8, 6.5, 6.2, 5.9, 5.9, 5.8, ~
## $ temp_grass     <dbl> 6.0, 6.3, 6.8, 6.2, 6.4, 3.9, 3.6, 3.4, 2.5, 2.5, 2.6, ~
## $ humidity       <dbl> 60, 63, 67, 69, 70, 76, 76, 77, 78, 71, 71, 72, 80, 81,~
## $ wind_direction <dbl> 47, 50, 52, 48, 45, 39, 30, 21, 4, 7, 27, 2, 344, 348, ~
## $ wind_speed     <dbl> 4.5, 6.0, 6.2, 6.4, 5.7, 5.4, 4.4, 3.5, 3.1, 3.1, 3.7, ~
## $ gust_direction <dbl> 53, 61, 54, 59, 44, 22, 31, 44, 7, 33, 45, 8, 346, 12, ~
## $ gust_speed     <dbl> 7.4, 10.4, 10.2, 10.2, 9.6, 8.8, 9.5, 8.7, 6.4, 7.5, 8.~
## $ rainfall_mm    <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ pressure_msl   <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ pressure_stn   <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ sunshine_hrs   <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
```

Two sources of NA were identified and confirmed as genuine data absence rather than processing errors: pressure data begins 20 days after the temperature record, and sunshine data begins on 12 April 2002, reflecting different sensor installation dates. A manual spot-check confirmed that pressure joins correctly for dates where both files overlap.

```r
# --- Filter to April 2002 ---

weather <- weather %>%
  filter(datetime_utc >= as.POSIXct("2002-04-12 01:00:00", tz = "UTC"))

nrow(weather)
```

```
## [1] 211616
```

The dataset was filtered to begin on 12 April 2002, the date from which sunshine data becomes available, ensuring all six variables are present across a consistent time window. The filtered dataset contains 252519 hourly observations spanning April 2002 to June 2026. A decision on whether to further restrict the analysis to the most recent 10 years (2016–2026) has been deferred to the EDA stage, where a distributional comparison between earlier and later periods will be performed to assess whether recent climate conditions differ meaningfully from the earlier record.

```r
# --- Convert UTC to NZ time and derive time columns ---
```

```r
weather <- weather %>%
  mutate(
    datetime_nzt = with_tz(datetime_utc, tzone = "Pacific/Auckland"),
    year   = year(datetime_nzt),
    month  = month(datetime_nzt, label = TRUE),
    hour   = hour(datetime_nzt),
    season = case_when(
      month(datetime_nzt) %in% c(12, 1, 2) ~ "Summer",
      month(datetime_nzt) %in% c(3, 4, 5)  ~ "Autumn",
      month(datetime_nzt) %in% c(6, 7, 8)  ~ "Winter",
      month(datetime_nzt) %in% c(9, 10, 11) ~ "Spring"
    )
  )
```

Timestamps were converted from UTC to New Zealand time using with_tz(). Season assignments follow the Southern Hemisphere calendar — summer spans December through February, autumn March through May, winter June through August, and spring September through November.

```r
# --- Save clean dataset ---

dir.create("Data/Clean", showWarnings = FALSE)
saveRDS(weather, "Data/Clean/dunedin_weather_clean.rds")
```

The clean dataset has been saved to Data/Clean/dunedin_weather_clean.rds and will be loaded at the start of the EDA document. Raw files remain unmodified."